Focus

Quantum Computing has Almost Arrived!

Quantum computing is slowly but surely transitioning from academic research to the real world. Once it comes into everyday use, its effects will be far-reaching, to say the least.

Quantum computing has rapidly transitioned from purely academic explorations to early industry deployments, with pilot projects emerging across finance, healthcare, and cybersecurity. Organisations are investing in proof-of-concepts for portfolio optimisation, molecular simulation, and quantum-safe encryption to prepare for post-quantum threats.

Conventional silicon-based computers consist of millions of transistors that switch between ‘on’ and ‘off’, corresponding to two possible values of bits -- 0 and 1. Processing relies on basic logic gates like AND, NOT and the like acting on individual bits or pairs of bits. Each bit’s state can be measured and distinguished without disturbing the system.

Quantum computers, by contrast, employ quantum bits or qubits which can occupy a superposition of both 0 and 1 at the same time. A two-qubit device can represent four states simultaneously, a three-qubit system eight states, a four-qubit system sixteen states, and an n-qubit system can register up to 2^n unique states in parallel. Qubits can also become entangled, meaning the state of one qubit is intrinsically linked to the state of another, which is a defining feature of quantum mechanics.

This inherent parallelism allows quantum machines to tackle complex problems far beyond the reach of classical architectures. Because qubit operations scale exponentially with the number of qubits, quantum computers promise dramatic speedups using the algorithms designed to exploit superposition and entanglement.

The best way to understand quantum computing is to contrast it with classical computing (Table 1).

The quantum computing market is projected to reach US$ 5 billion by 2030, with fault-tolerant systems expected to become commercially viable in the early 2030s, driving innovation and collaboration across sectors. IBM and AMD have joined forces to create scalable, fault-tolerant quantum computing platforms combining IBM’s quantum expertise with AMD’s supercomputing strengths.

The European Innovation Council-funded QCDC (Quantum Computers for Data Centers) project has successfully